PROBABILISTIC LEARNERS

Naive Bayes and Discriminant Analysis

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The Big Picture

Is This Person a Librarian?

Ask yourself these questions:

1. How common are librarians in the first place? (Base rate)
2. How probable are the features in the picture given that someone is a librarian?
   - The book stack
   - The stern look
   - The glasses
   - The shushing . . .

Classification as Belief Updating

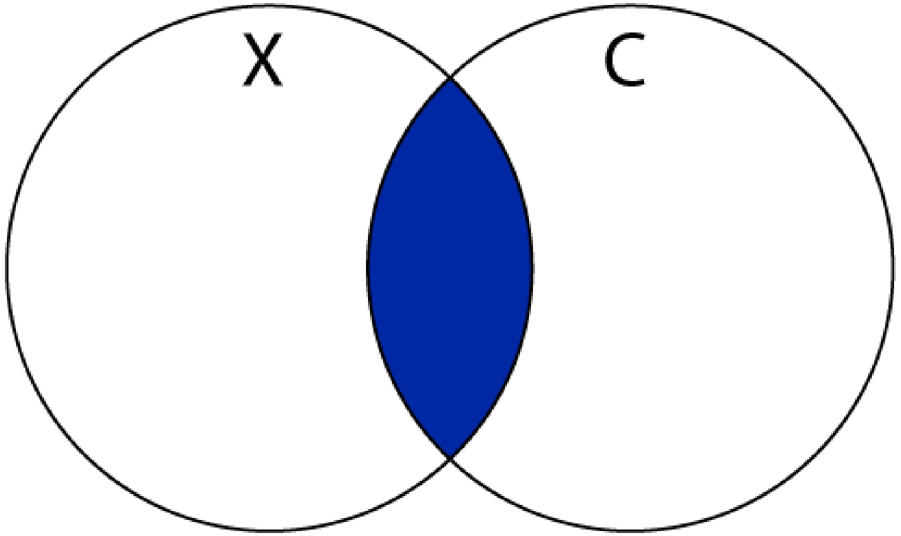
- Treat the classification as a Bayesian belief-updating problem.
- We have a **prior**—the base rate.
- We have **data**—the attributes shown in the image.
- We can now derive a **posterior** for each of a set of classes.

Probabilistic Learning

- Probabilistic learners apply the rules of probability calculus to form predictions.
- They are typically applied to classification tasks.
- They return probabilities in lieu of class labels.

Bayes' Theorem

Some Rules of Probability Calculus



- The joint probability is the probability that events X and C both hold true.
- This can be written as

$$\mathbf{Pr}(X \cap C) = \mathbf{Pr}(C|X) \cdot \mathbf{Pr}(X)$$

- We can also write

$$\mathbf{Pr}(C|X) = \frac{\mathbf{Pr}(X \cap C)}{\mathbf{Pr}(X)}$$

The Bayesian Insight

Bayes' theorem states that

$$\Pr(C|X) = \frac{\Pr(X|C) \cdot \Pr(C)}{\Pr(X)} \qquad (1)$$

where:

- $\Pr(C)$ is the prior probability of observing C.
- $\Pr(X|C)$ is the likelihood (data).
- $\Pr(X)$ is the marginal likelihood.
- $\Pr(C|X)$ is the posterior probability.

Thomas Bayes (1701-1761)

The Bayesian Kernel

- $\mathbf{Pr}(X)$ is a normalizing constant that we can sometimes ignore.
- In that case, Bayes' theorem can also be written as $\mathbf{Pr}(C|X) \propto \mathbf{Pr}(X|C) \cdot \mathbf{Pr}(C)$.
- We shall use this shortcut because it simplifies things.
- It suffices if the goal is to predict a class; for class *probabilities*, $\mathbf{Pr}(X)$ has to be taken into account.

Application to Classification

- Let $\mathbf{Pr(Librarian)}$ be the prior probability of encountering a librarian (the base rate).
- Let $X = \mathbf{Books}$ and let $\mathbf{Pr(Books|Librarian)}$ be the probability of seeing books in the vicinity of a librarian.
- Then,
  $\mathbf{Pr(Librarian|Books)} \propto \mathbf{Pr(Books|Librarian)} \cdot \mathbf{Pr(Librarian)}$.
- This is the foundation of the naive Bayes classifier.

The Naive Bayes Classifier for Discrete Features

A First Example with Discrete Predictive Features

	HDI Low	HDI Medium	HDI High	HDI Very High
Authoritarian	14	14	12	9
Hybrid	11	11	6	0
Flawed Democracy	1	7	9	17
Full Democracy	0	0	0	16

- We are interested in the class "Flawed Democracy".
- We observe a high HDI.
- $\mathbf{Pr(Flawed\ Dem) = 47/123 = 0.276}$.
- $\mathbf{Pr(HDI\ High|Flawed\ Dem) = 0.265}$.
- Then $\mathbf{Pr(Flawed\ Dem|HDI\ High) \propto 0.265 \cdot 0.276 \propto 0.073}$.
- Moreover, $\mathbf{Pr(Autho|HDI\ High) \propto 0.098}$, $\mathbf{Pr(Hybrdid|HDI\ High) \propto 0.049}$, and $\mathbf{Pr(Full\ Dem|HDI\ High) \propto 0.000}$.

MAP

The class prediction is based on the **maximum a posteriori** (MAP) probability:

$$
\begin{aligned}
\mathrm{MAP} &= \max_j \Pr(y = j|\boldsymbol{x}) \\
&= \max_j \frac{\Pr(\boldsymbol{x}|y = j) \cdot \Pr(y = j)}{\Pr(\boldsymbol{x})} \\
&= \max_j \Pr(\boldsymbol{x}|y = j) \cdot \Pr(y = j)
\end{aligned}
\tag{2}
$$

Illustration

- In our example, the MAP is for authoritarian regimes $\rightarrow$ predicted label.
- The full set of posterior probabilities are:
  - $$\Pr(\text{Autho}|\text{HDI High}) = 0.444$$
  - $$\Pr(\text{Hybrid}|\text{HDI High}) = 0.222$$
  - $$\Pr(\text{Flawed Dem}|\text{HDI High}) = 0.333$$
  - $$\Pr(\text{Full Demo}|\text{HDI High}) = 0.00$$.

Multiple Discrete Features

	Low Literacy				High Literacy			
HDI ->	**L**	**M**	**H**	**VH**	**L**	**M**	**H**	**VH**
A	18	12	5	1	0	4	9	8
H	11	12	1	0	0	0	7	2
Flawed	1	6	3	1	0	1	10	21
Full	0	0	0	1	0	1	0	5

- So far $\boldsymbol{x} = x$, i.e., a single feature.
- Typically, $\boldsymbol{x}$ is of dimensionality $P \gg 1$.
- Then, the procedure outlined so far would entail a multi-way cross-tabulation between all of the features and the labels.
- This results in many cells, many of which will be empty.

The Simple Beauty of Naivety

- Let us assume that all predictive features are statistically independent.
- In this case,

$$\begin{aligned}\Pr(y = j|\boldsymbol{x}) &\propto \Pr(y = j) \cdot \Pr(\boldsymbol{x}|y = j)\\ &\propto \Pr(y = j) \prod_{i=1}^{P} \Pr(x_i|y = j)\end{aligned}$$

- We call this the **naive Bayes classifier**.
- It is computationally simple and, despite the unrealistic assumption, often remarkably good at prediction.

A Common Complication

	Low	Medium	High	Very High
A	14	14	12	9
H	11	11	6	0
Flawed	1	7	9	17
Full	0	0	0	16

Empty cells can result in counter-intuitive results:

- All features point to full democracy.
- We now add a feature that produces an empty cell for full democracies.
- Suddenly it is impossible to classify an instance as a full democracy

Laplacian Smoothing

- We turn the zero-conditional probability into a small positive number.
- We compute

$$q_{ij} = \frac{f_{ij} + \alpha}{n_{1j} + \alpha \cdot M}$$

  where

  1. f_{ij} is the number of instances with value i on x and $y = j$.
  2. n_{1j} is the number of training instances with class $y = j$.
  3. $\alpha \geq 0$ is a smoothing parameter.
  4. M is the number of classes.

- Often we choose $\alpha = 1$, which results in add-one smoothing.
- More typically, we tune α.

Bypassing Naivety

- There is work that seeks to alleviate independence through weighting (Zaidi et al. 2013).
- Implementation in `tidymodels` is not straightforward, but there is a package: `bnclassify`.
- Given that naive Bayes often makes very good predictions, I have not yet seen many applications of the updated classifier.

The Naive Bayes Classifier for Continuous Features

Going Continuous

With numeric predictive features, we can write the naive Bayes classifier as

$$\Pr(y = j|\boldsymbol{x}) \propto \Pr(y = j) \prod_{i=1}^{P} p(x_i|y = j)$$

where $p(.)$ denotes a probability density function or pdf.

Going Gaussian

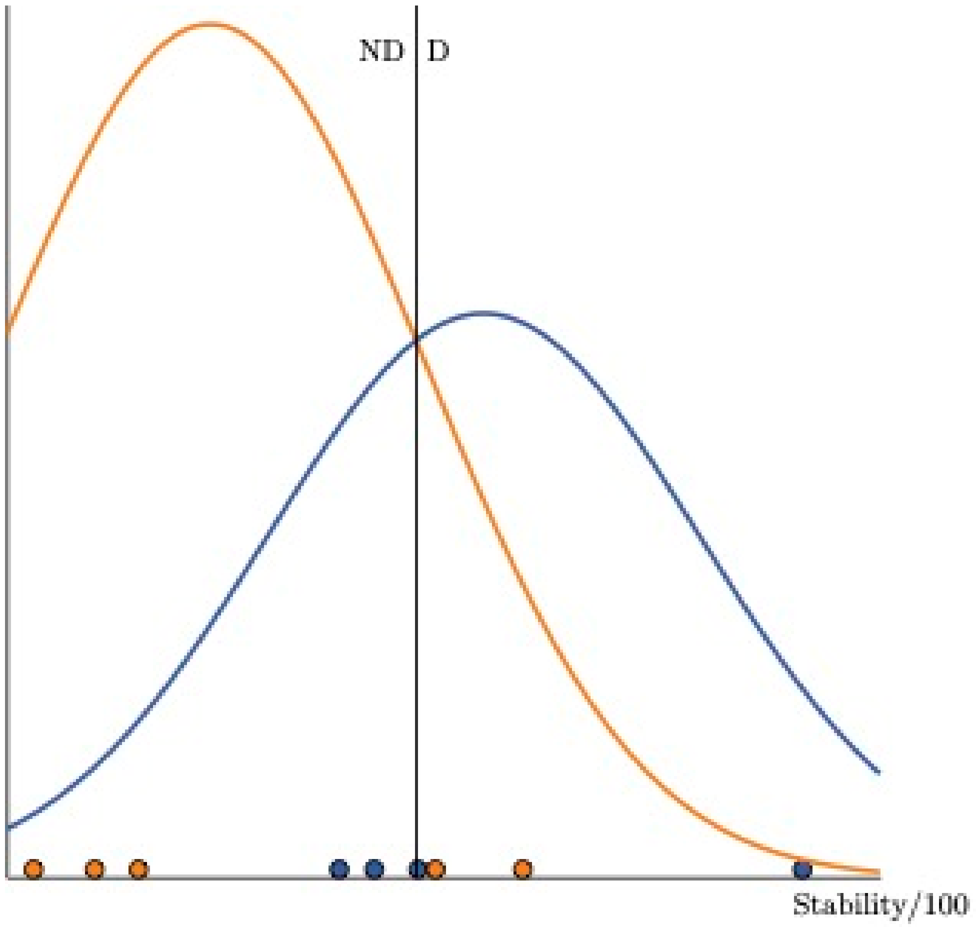
Sometimes, the pdf is specified to be the normal or Gaussian pdf:

$$p(x_i|y=j) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} \exp\left\{-\frac{1}{2}\frac{(x_i-\mu_{ij})^2}{\sigma_{ij}^2}\right\}$$

where

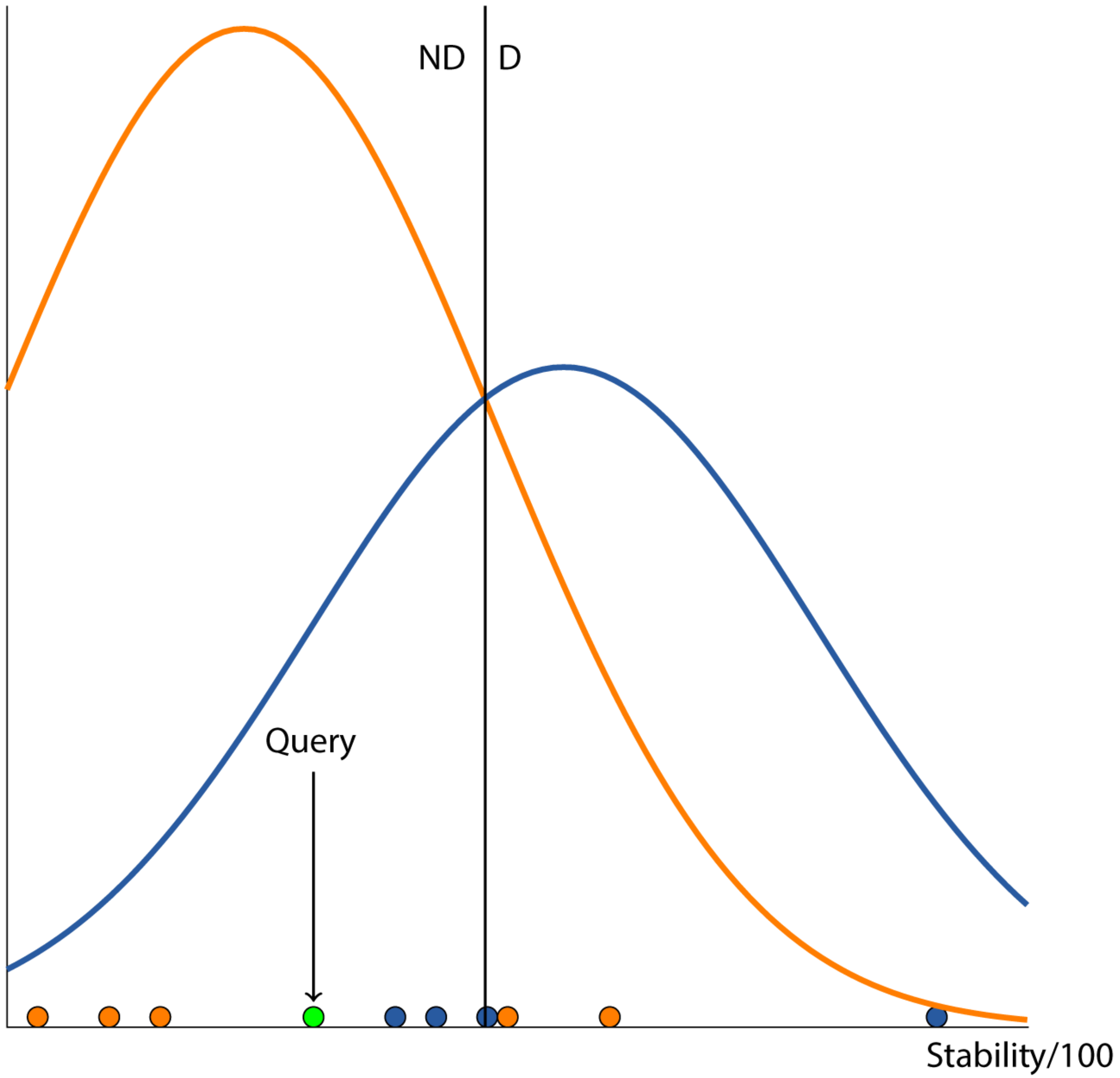
- μ_{ij} is the mean of feature i in class j.
- σ_{ij} is the standard deviation of feature i in class j.

Decision Boundary



The decision boundary is a hyper-surface that partitions the feature space into a set of classes.

Prediction



Using Kernels

- Not everyone likes to use the normal distribution.
- Two liabilities are:
  - → This prevents us from introducing categorical predictive features
  - → Many predictive features simply are not normally distributed
- An alternative is to rely on kernel density estimates of the predictive features.
- The kernel smoothness (bandwidth) is another tuning parameter.

Setting Up Naive Bayes in `tidymodels`

Data | Split | Recipe | Model+Flow | Grid | CV

```r
library(rio)
library(tidyverse)
world22_df <- import("Data/world22.xlsx")
work_df <- tibble(world22_df) %>%
  column_to_rownames(var = "COUNTRY") %>%
  filter(!is.na(REGIME)) %>%
  mutate(DEMOCRACY = ifelse(REGIME=="Flawed Democracy"|
                              REGIME=="Full Democracy", 1, 2),
         DEMOCRACY = factor(DEMOCRACY, labels = c("Democracy", "Other")),
         LOGPCGDP = log10(GDPPERCAP)) %>%
  select(DEMOCRACY, POP21, LITERACY, HDI21, LOGPCGDP, GDPGROWTH, GINI,
         EFIDX, STABILITY, GOVEFF, RULEOFLAW)
```

Tuning Naive Bayes in `tidymodels`

Metric | Results | Plot | Summary | Optimal

```
bayes_metrics <- metric_set(roc_auc)
```

Prediction with Naive Bayes in `tidymodels`

Finalize | Predict

```
bayes_updated <- finalize_model(bayes_spec,
                                select_best(bayes_tune))
workflow_new <- initial_flow %>%
  update_model(bayes_updated)
new_fit <- workflow_new %>%
  fit(data = demo_train)
```

Performance with Multiple Categories

Log-Loss

- Imagine we have M categories that are being predicted.
- For each category j and each instance i, we obtain a predicted probability π_{ij} .
- We can now define a **log-loss** performance metric (a.k.a. cross-entropy):

$$\mathrm{lnL} = -\sum_{i=1}^{n_2}\sum_{j=1}^{M} y_{ij} \cdot \ln \pi_{ij}$$

- The lowest value this can take is 0.
- Sometimes, the log-loss is converted to a *mean* log-loss through division by n_2.

The Tribulations of a Betting Person

- Imagine we predict membership in $\boldsymbol{j}$ with $\boldsymbol{\pi_j = 1}$; we are certain of our guess. If $\boldsymbol{y = j}$ and $\boldsymbol{y_j = 1}$, then lnL is 0. Perfect!
- Had we just barely favored $\boldsymbol{j}$, e.g., by setting $\boldsymbol{\pi_j = 0.501}$, then we would have incurred a loss: lnL = 0.691.
- However, it also works the other way around:
  - → We are certain of $\boldsymbol{j}$ but in actuality $\boldsymbol{k}$ occurs $\Rightarrow$ high loss
  - → We are barely opting for $\boldsymbol{j}$ and in actuality $\boldsymbol{k}$ occurs $\Rightarrow$ lower loss

Discriminant Analysis

Generalizing Gaussian Naive Bayes

- Assume the predictive features follow the $\boldsymbol{P}$-variate normal distribution

$$p(\boldsymbol{x}|y=j) = \frac{\exp\left[-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_j)^\top \boldsymbol{\Sigma}_j^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_j)\right]}{\sqrt{(2\pi)^P|\boldsymbol{\Sigma}_j|}}$$

where

→ $\boldsymbol{\mu}_j$ is a vector of feature means for class $\boldsymbol{j}$.

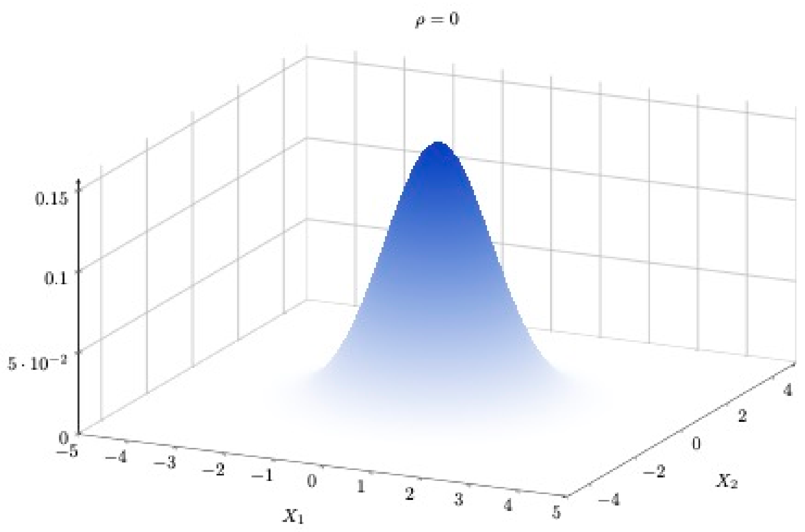
→ $\boldsymbol{\Sigma}_j$ is the variance-covariance matrix for the features in class $\boldsymbol{j}$.

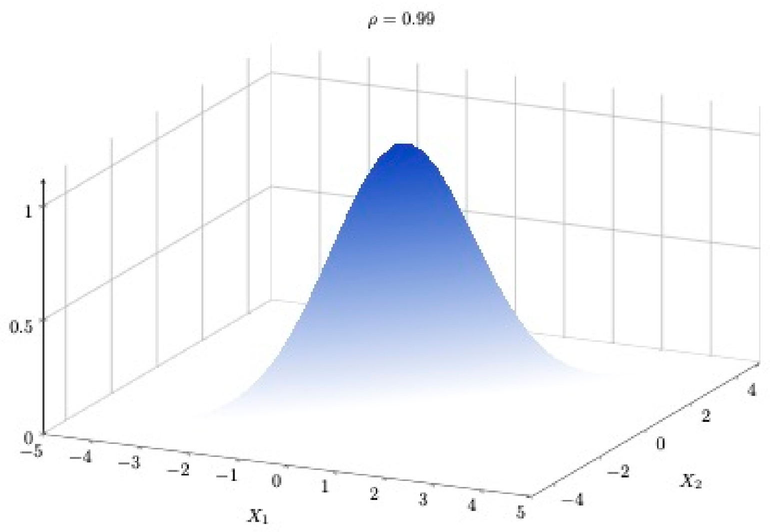
→ $|.|$ denotes the determinant.

Generalizing Gaussian Naive Bayes Cont'd

- Gaussian naive Bayes is the special case, where we assume all covariances among the features to be 0.
- However, there are other models making different assumptions; these generally go under the name of **discriminant analysis**.

Multivariate Normality





Overview

Method	Means	VC-Matrix	Parameters
Gaussian Naive Bayes	Vary by class	Diagonal	$2MP + M - 1$
Linear Discriminant	Vary by class	Constant across classes (i.e., pooled)	$MP + .5P(P + 1) + M - 1$
Quadratic Discriminant	Vary by class	Vary by class	$MP + .5MP(P + 1) + M - 1$

Linear Discriminant Method

- The idea goes back to Fisher (1936).
- The posterior probability of instance $\boldsymbol{i}$ belonging to class $\boldsymbol{j}$ is

$$\mathbf{Pr}(y_i = j|\boldsymbol{x}_i) \propto \mathbf{Pr}(y_i = j) \cdot (2\pi)^{-.5P} \cdot |\boldsymbol{\Sigma}|^{-.5} \cdot \exp\left[-\frac{1}{2}(\boldsymbol{x}_i - \boldsymbol{\mu}_j)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i - \boldsymbol{\mu}_j)\right]$$

- The class means and pooled variance-covariance matrix are estimated using maximum likelihood.
- Classes are determined based on the MAP.

What Is Linear about This?

- The decision boundary between classes $\boldsymbol{j}$ and $\boldsymbol{k}$ is $\mathbf{Pr}(y = j|\boldsymbol{x}) = \mathbf{Pr}(y = k|\boldsymbol{x})$ or

$$\ln\left(\frac{\mathrm{Pr}(y = j|\boldsymbol{x})}{\mathrm{Pr}(y = k|\boldsymbol{x})}\right) = 0$$

- Substituting the posterior formulas and rearranging terms yields

$$(\boldsymbol{\mu}_j - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{x} = c + \frac{1}{2}(\boldsymbol{\mu}_j^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_j - \boldsymbol{\mu}_k^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k)$$

  where $\boldsymbol{c}$ is a constant based on the differences in the log-priors.

- This is of the form $\boldsymbol{b}^\top \boldsymbol{x} = \boldsymbol{a}$ or a linear function.

Prediction Using Linear Scores

- We know that $\ln \Pr(y = j|\boldsymbol{x}) \propto \ln \Pr(y = j) + \ln p(\boldsymbol{x}|y = j)$.
- With some math, the right-hand side becomes

$$\ln \Pr(y = j) - \frac{P}{2}\ln \pi - \frac{1}{2}\ln|\boldsymbol{\Sigma}| - \frac{1}{2}\boldsymbol{\mu}_j^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_j + \boldsymbol{\mu}_j^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{x}$$

- Dropping all of the terms that do not vary in $\boldsymbol{j}$, we obtain the **linear score** function

$$\boldsymbol{s}_j = -\frac{1}{2}\boldsymbol{\mu}_j^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_j + \boldsymbol{\mu}_j^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{x} + \ln \Pr(y = j)$$

- We assign to whichever category has the highest score.
- The term $\boldsymbol{d}_j^\top = \boldsymbol{\mu}_j^\top \boldsymbol{\Sigma}^{-1}$ is known as the vector of **discriminant parameters**.

Fitting a Linear Discriminant Model in `tidymodels`

Data | Split | Recipe | Model | Fit | Res 1 | Res 2 | Res 3

```r
work_df <- tibble(world22_df) %>%
  column_to_rownames(var = "COUNTRY") %>%
  filter(!is.na(REGIME)) %>%
  mutate(REGIME = as.factor(REGIME),
         LOGPCGDP = log10(GDPPERCAP)) %>%
  select(REGIME, HDI21, LOGPCGDP, GINI, EFIDX, RULEOFLAW)
```

Evaluating LDA in `tidymodels`

```r
regime_test <-
  predict(regime_fit, regime_test, type = "prob") %>%
  bind_cols(regime_test)
regime_test %>%
  mn_log_loss(REGIME, 1:4)
```

```
# A tibble: 1 × 3
  .metric     .estimator .estimate
  <chr>       <chr>          <dbl>
1 mn_log_loss multiclass     0.841
```

Quadratic Discriminant Analysis

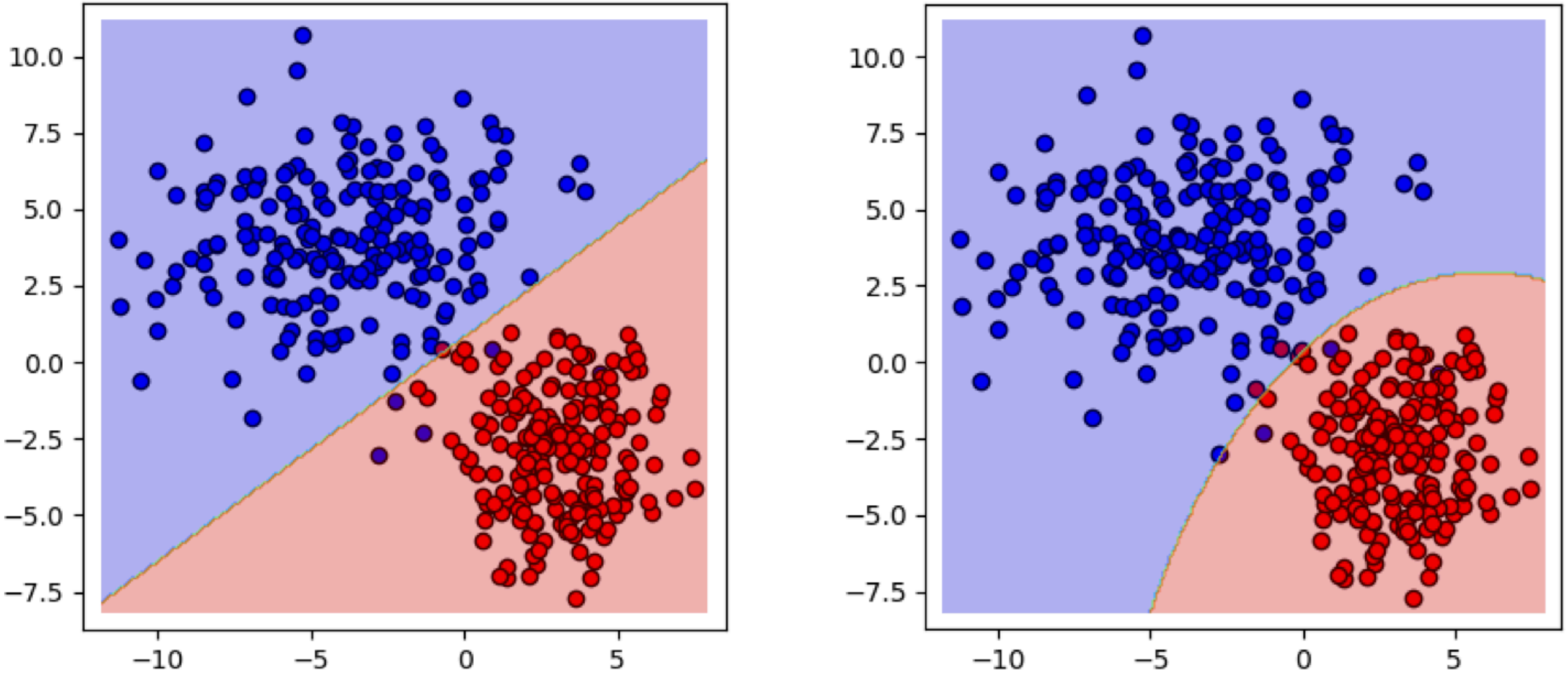
- In quadratic discriminant analysis, we allow both the means and variance-covariance matrices to vary by class.
- This results in a quadratic boundary function:

$$(\boldsymbol{x} - \boldsymbol{\mu}_j)^\top \boldsymbol{\Sigma}_j^{-1}(\boldsymbol{x} - \boldsymbol{\mu}_j) - (\boldsymbol{x} - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1}(\boldsymbol{x} - \boldsymbol{\mu}_k) = k$$

- Here,

$$k = 2\left[\ln \Pr(y = j) - \ln \Pr(y = k)\right] + \ln|\boldsymbol{\Sigma}_k| - \ln|\boldsymbol{\Sigma}_j|$$

Comparison with LDA



Source: Ghojogh and Crowley (2019, 11)

QDA in tidymodels

Model+Fit | Results | Prediction

```
library(discrim)
qda_spec <- discrim_quad(regularization_method = NULL) %>%
  set_mode("classification") %>%
  set_engine("MASS")
qda_flow <- workflow() %>%
  add_recipe(lda_recipe) %>%
  add_model(qda_spec)
regime_fit <- fit(qda_flow, regime_train)
```

References

Fisher, Ronald A. 1936. “The Use of Multiple Measurements in Taxonomic Problems.” *Annals of Eugenics* 7 (2): 179–88.

Ghojogh, Benyamin, and Mark Crowley. 2019. “Linear and Quadratic Discriminant Analysis: Tutorial.” https://doi.org/10.48550/ARXIV.1906.02590.

Zaidi, Nayyar A., Jesús Cerquides, Mark J. Carman, and Geoffrey I. Webb. 2013. “Alleviating Naive Bayes Attribute Independence Assumption by Attribute Weighting.” *Journal of Machine Learning Research* 14 (24): 1947–88.